Functional Programming and Verification revision course

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Outline

Functional Programming and Haskell

Types

Proofs

Correctness

I/O

Evaluation

Time complexity analysis

Plan

Functional Programming and Haskell

- Basic Haskell
- Recursion, guards, pattern matching
- List comprehensions
- QuickCheck
- Polymorphism
- Currying, partial application, higher-order functions

Basic Haskell

function types `f :: a -> b -> c`

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prefix/infix precedence	`f a ‘g‘ b` means `(f a) ‘g‘ b`
$ sign	`f $ a ‘g‘ b` means `f (a ‘g‘ b)`

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`Integer`	unbounded integers
`Char`	`'a'`
`String`	`"hello" :: [Char]`
`(a,b)` (Tuple)	`("hello",1) :: (String,Int)`

Tuples

```
(1,"hello") :: (Int,String)
(x,y,z) :: (a,b,c)
-- ...
```

Prelude functions: `fst`, `snd`

Lists

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Cons `(:)` and `[]` are constructors of lists, that is a function that uniquely constructs a value of the list type.

Intuitively: `(:) :: a -> [a] -> [a]`.

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`unzip :: [(a,b)] -> ([a],[b])`	separate list of tuples into list of components

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search for functions by type signature on `https://hoogle.haskell.org/`.

Ranges

```
[1..5]
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[1..5]
= [1,2,3,4,5]

[1,3..10]
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[1..]
= [1,2,3...]

[1,3..]
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Local definitions

`let x =` e_1 `in` e_2

defines `x` locally in e_2.

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`let x =` e_1 `in` e_2

defines `x` locally in e_2.

e_2 `where x =` e_1

also defines `x` locally in e_2 where e_2 has to be a function definition.

Recursion, guards, pattern matching

Guards

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Example: maximum of two integers.

```
max2 :: Integer -> Integer -> Integer
max2 x y
```

Recursion, guards, pattern matching

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Example: maximum of two integers.

```
max2 :: Integer -> Integer -> Integer
max2 x y
  | x >= y    = x
  | otherwise = y
```

Recursion

Reduce problem into a solving a series of smaller problems of a similar kind.

Example

```
factorial :: Integer -> Integer
factorial n
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```
factorial :: Integer -> Integer
factorial n
  | n == 0 = 1                          -- base case
  | n > 0  = n * factorial (n - 1)      -- recursive case
```

Accumulating parameter

Alternatively, `factorial` could be defined as

```
factorial :: Integer -> Integer
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  where
    aux :: Integer -> Integer -> Integer
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The resulting function is tail recursive, that is the recursive call is located at the very end of its body.
Therefore, no computation is done after the recursive function call returns.

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In general, recursion using accumulating parameters is less readable.

Pattern matching

A more compact syntax for recursion:

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factorial 0 = 1
factorial n | n > 0 = n * factorial (n - 1)
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Patterns are expressions consisting only of constructors, variables, and literals.

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Examples

```
head :: [a] -> a
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```
head :: [a] -> a
head (x : _) = x

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null :: [a] -> Bool
null []       = True
null (_ : _) = False
```

Constructors vs Types

What is the difference between `True` and `Bool`?

- `True` is a constructor, `Bool` is a type.
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Constructor?

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Constructor?

- `False` yes
- `(:)`

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Constructor?

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- `(:)` yes
- `Maybe`

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- `Just` yes
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Case

Pattern matching in nested expressions

```
singleOrEmpty :: [a] -> Bool
singleOrEmpty xs = case xs of []  -> True
                              [_] -> True
                              _   -> False
```

List comprehensions

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- a test is a Boolean expression

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[(x, even x) | x <- [1..3]]
= [(1, False), (2, True), (3, False)]
```

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Generators are reduced from left to right.
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= [(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)]
```

The meaning of list comprehensions

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[e | x <- [a1,...,an]]
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Run `quickCheck prop_max2` from GHCI to check the property.

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`length :: [a] -> Int` is defined for all types `a`
where `a` is a type variable.

Subtype vs parametric polymorphism

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  types may contain universally quantified type variables that are then replaced by actual types.
- subtype polymorphism
  any object of type `T'` where `T'` is a subtype of `T` can be used in place of objects of type `T`.

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Haskell uses parametric polymorphism.

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Type variables can be constrained by type constraints.

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- `Fractional`
- `Ord`
- `Eq`

Type constraints

Type variables can be constrained by type constraints.

```
(+) :: Num a => a -> a -> a
```

Function `(+)` has type `a -> a -> a` for any type `a` of the type class `Num`.

Some type classes:

- `Num`
- `Integral`
- `Fractional`
- `Ord`
- `Eq`
- `Show`

Quiz

```
f x y z = if x then y else z
```

Quiz

```
f x y z = if x then y else z
f :: Bool -> a -> a -> a

f x y = [(x,y), (y,x)]
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f x y z = if x then y else z
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f :: a -> a -> [(a,a)]

f x = [length u + v | (u,v) <- x ]
```

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f :: a -> a -> [(a,a)]

f x = [length u + v | (u,v) <- x ]
f :: [([a],Int)] -> [Int]

f x y = [u ++ x | u <- y, length u < x ]
```

Quiz

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invalid

f x y = [[(u,v) | u <- w, u, v <- x] | w <- y]
```

Quiz

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f x y z = if x then y else z
f :: Bool -> a -> a -> a

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f x y = [u ++ x | u <- y, length u < x ]
invalid

f x y = [[(u,v) | u <- w, u, v <- x] | w <- y]
f :: [a] -> [[Bool]] -> [[(Bool, a)]]
```

Currying

A function is curried when it takes its arguments one at a time, each time returning a new function.

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Example

```
f :: Int -> Int -> Int
f x y = x + y
```

```
f :: Int -> (Int -> Int)
f x = \y -> x + y
```

Currying

A function is curried when it takes its arguments one at a time, each time returning a new function.

Example

```
f :: Int -> Int -> Int  f :: Int -> (Int -> Int)
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f a b                   (f a) b
```

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= a + b
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= (\y -> a + y) b
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= a + b
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f :: Int -> (Int -> Int)
f x = \y -> x + y

(f a) b
= (\y -> a + y) b
= a + b
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Any function of two arguments can be viewed as a function of the first argument that returns a function of the second argument.

Anonymous functions (lambdas)

An anonymous function (or lambda abstraction) is a function without a name.

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Examples

```
(\x -> x + 1) 4
```

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= 5

(\x y -> x + y) 3 5
```

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(\x y -> x + y) 3 5
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What is the type of `\n -> iter n succ` where
`iter :: Integer -> (a -> a) -> (a -> a)`
`succ :: Integer -> Integer`

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`Integer -> (Integer -> Integer)`

Partial application

Every function of n parameters can be applied to less than n arguments.

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Expressions of the form (*infixop expr*) or (*expr infixop*) are called sections.

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- `(.) :: (b -> c) -> (a -> b) -> (a -> c)`
- `const :: a -> (b -> a)`

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- `(.) :: (b -> c) -> (a -> b) -> (a -> c)`
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- `filter :: (a -> Bool) -> [a] -> [a]`

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- `filter :: (a -> Bool) -> [a] -> [a]`
- `map :: (a -> b) -> [a] -> [b]`
- `all, any :: (a -> Bool) -> [a] -> Bool`

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- `(.) :: (b -> c) -> (a -> b) -> (a -> c)`
- `const :: a -> (b -> a)`
- `curry :: ((a,b) -> c) -> (a -> b -> c)`
- `uncurry :: (a -> b -> c) -> ((a,b) -> c)`
- `filter :: (a -> Bool) -> [a] -> [a]`
- `map :: (a -> b) -> [a] -> [b]`
- `all, any :: (a -> Bool) -> [a] -> Bool`
- `takeWhile, dropWhile :: (a -> Bool) -> [a] -> [a]`

Fold

Folding is the most elementary way
of combining elements of a list.

Right-associative (`foldr`):

```
foldr :: (b -> a -> a) -> a -> [b] -> a
foldr f a []     = a
foldr f a (x:xs) = f x (foldr f a xs)
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= 1 + (2 + (3 + 0))
= 1 + (2 + 3)
= 1 + 5 = 6
```

Plan

- Types
  - Type aliases
  - Type Classes
  - Algebraic Data Types
  - Modules, Abstract Data Types
  - Type inference

Type aliases

Allows the renaming of a more complex type expression.

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Examples

```
type String = [Char]
type List a = [a]
```

Type Classes

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1. creating a type class $\sim$ creating an interface (define set of functions)
2. instantiating a type class $\sim$ implementing an interface (implement a set of functions for a member of a type class)

Type Classes

Examples

```
class Eq a where
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```

Constrained instances

Instances of type classes can be constrained.

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```
instance (Eq a) => Eq [a] where
  [] == [] = True
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  _ == _ = False
```

Subclasses

Example

```
class (Eq a) => Ord a where
  (<=), (<), (>=), (>) :: a -> a -> Bool
```

Class `Ord` inherits all functions of class `Eq`.

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Before instantiating a subclass with a type, the type must be an instance of all *"superclasses"*.

Subclasses

Example

```
class (Eq a) => Ord a where
  (<=), (<), (>=), (>) :: a -> a -> Bool
```

Class `Ord` inherits all functions of class `Eq`.

Before instantiating a subclass with a type, the type must be an instance of all *"superclasses"*.

```
instance Ord Bool where
  b1 <= b2 = not b1 || b2
  b1 < b2 = b1 <= b2 && not(b1 == b2)
```

Algebraic Data Types

A custom datatype with one or more constructors.

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$$\texttt{data}\ \mathit{type}\ a_1 \ldots a_n\ \texttt{=}\ \mathit{constructor}\ a_k \ldots a_l\ \texttt{|}\ \ldots$$

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Constructors are

- a *prefix operator* starting with a capital letter; or
- an *infix operator* starting with `:`.

Algebraic Data Types

Examples

```
data Bool = False | True
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Algebraic Data Types

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data Nat = Zero | Suc Nat
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data [a] = [] | (:) a [a]
  deriving Eq

data Tree a = Empty | Node a (Tree a) (Tree a)
  deriving (Eq, Show)
```

Algebraic Data Types

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Terminology:

- a *n*-ary constructor is a function that unambiguously constructs values of a type encapsulating *n* arguments.
- nullary constructors are also called constants.
- a type that expects a *type argument* is called a *parametrized type*.
- data constructors are used at the *term level*, type constructors are used at the *type level*.

Algebraic Data Types

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A datatype can be thought of as the set of possible values of that type.

- the cardinality of a datatype is the number of all its possible values.
- a sum type is a type with more than one constructor (similar to a logical $\vee$).
- a product type is a type whose data constructor takes more than one argument (similar to a logical $\wedge$).

Pattern matching

Pattern matching works just the same for custom constructors as for predefined constructors.

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Examples

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find :: Ord a => a -> Tree a -> Bool
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find :: Ord a => a -> Tree a -> Bool
find _ Empty = False
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find :: Ord a => a -> Tree a -> Bool
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find x (Node a l r)
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  | x < a     = find x l
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Examples

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find :: Ord a => a -> Tree a -> Bool
find _ Empty = False
find x (Node a l r)
  | x < a     = find x l
  | a < x     = find x r
  | otherwise = True

insert :: Ord => a -> Tree a -> Tree a
```

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  | otherwise = True

insert :: Ord => a -> Tree a -> Tree a
insert x Empty = Node x Empty Empty
```

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find _ Empty = False
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  | otherwise = True

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insert x (Node a l r)
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insert x Empty = Node x Empty Empty
insert x (Node a l r)
  | x < a     = Node a (insert x l) r
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  | otherwise = True

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insert x (Node a l r)
  | x < a     = Node a (insert x l) r
  | a < x     = Node a l (insert x r)
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```

Modules

Collection of type, function, class and other definitions.

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Examples

`module M where`
exports everything defined in `M`

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`module M where`
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`module M (T, f, ...) where`
exports only `T, f, ...`

Exporting data types

```
module M (T) where
data T = ...
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exports only `T` but not its constructors

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module M (T) where
data T = ...
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```
module M (T(C,D,...)) where
data T = ...
```

exports `T` and its constructors `C, D, ...`

Exporting data types

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module M (T) where
data T = ...
```

exports only `T` but not its constructors

```
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data T = ...
```

exports `T` and its constructors `C, D, ...`

```
module M (T(..)) where
data T = ...
```

exports `T` and all its constructors

Exporting data types

```
module M (T) where
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exports only `T` but not its constructors

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Not allowed (why?):

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Not allowed (why?):

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```

Constructors could have the same name as a type.

Abstract Data Types

Hides data representation by wrapping data in a constructor that is not exported.

Abstract Data Types

Example

```
module Set (Set, empty, insert, isin, size) where
-- Interface
empty :: Set a
insert :: Eq a => a -> Set a -> Set a
isin :: Eq a => a -> Set a -> Bool
size :: Set a -> Int
```

Abstract Data Types

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newtype Set a = Set [a]
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size :: Set a -> Int
-- Implementation
newtype Set a = Set [a]

empty = Set []
insert x (Set xs) = Set (if elem x xs then xs else x:xs)
```

Abstract Data Types

Example

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empty :: Set a
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module Set (Set, empty, insert, isin, size) where
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size :: Set a -> Int
-- Implementation
newtype Set a = Set [a]

empty = Set []
insert x (Set xs) = Set (if elem x xs then xs else x:xs)
isin x (Set xs) = elem x xs
size (Set xs) = length xs
```

`type` vs `data` vs `newtype`

- `type` is used to create type aliases

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- `data` is used to create algebraic data types (types witha custom shape)

`type` vs `data` vs `newtype`

- `type` is used to create type aliases
- `data` is used to create algebraic data types (types witha custom shape)
- `newtype` is used to create a custom constructor for a single type without adding any runtime overhead

Type inference

Inferring/reconstructing the type of an expression.

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Given an expression e.

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Type inference

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Given an expression e.

1. give all variables in e distinct type variables
2. give each function $f :: T$ in e a new general type with fresh type variables
3. for each sub-expression in e set up an equation linking the type of parameters and arguments
4. simplify the set of equations by replacing equivalences

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Type inference

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Step 1

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Step 1

1. `u :: a`
2. `v :: b`

Step 2

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Step 1

1. `u :: a`
2. `v :: b`

Step 2

1. `head :: [c] -> c`

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Step 1

1. `u :: a`
2. `v :: b`

Step 2

1. `head :: [c] -> c`
2. `concat :: [[d]] -> [d]`

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Step 1

1. `u :: a`
2. `v :: b`

Step 2

1. `head :: [c] -> c`
2. `concat :: [[d]] -> [d]`
3. `last :: [e] -> e`

Type inference

Example

Given `f u v = min (head u) (last (concat v))`

Step 1

1. `u :: a`
2. `v :: b`

Step 2

1. `head :: [c] -> c`
2. `concat :: [[d]] -> [d]`
3. `last :: [e] -> e`
4. `min :: Ord f => f -> f -> f`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`

Step 3

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`

Step 3

1. from `head u` derive `[c] = a`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`

Step 3

1. from `head u` derive `[c] = a`
2. from `concat v` derive `[[d]] = b`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`

Step 3

1. from `head u` derive `[c] = a`
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3. from `last (concat v)` derive `[e] = [d]`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`

Step 3

1. from `head u` derive `[c] = a`
2. from `concat v` derive `[[d]] = b`
3. from `last (concat v)` derive `[e] = [d]`
4. from `min (head u) (last (concat v))` derive `f = c` and `f = e`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update
   - `u :: [c]`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update
   - `u :: [c]`
2. apply `[[d]] = b` and update

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

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1. apply `[c] = a` and update
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Given `f u v = min (head u) (last (concat v))`
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1. apply `[c] = a` and update
   - `u :: [c]`
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   - `v :: [[d]]`
3. apply `[e] = [d]` to get `e = d` and update

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update
   - `u :: [c]`
2. apply `[[d]] = b` and update
   - `v :: [[d]]`
3. apply `[e] = [d]` to get `e = d` and update
   - `v :: [[e]]`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update
   - `u :: [c]`
2. apply `[[d]] = b` and update
   - `v :: [[d]]`
3. apply `[e] = [d]` to get `e = d` and update
   - `v :: [[e]]`
   - `concat :: [[e]] -> [e]`

Type inference

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Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

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Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

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1. apply `[c] = a` and update
   - `u :: [c]`
2. apply `[[d]] = b` and update
   - `v :: [[d]]`
3. apply `[e] = [d]` to get `e = d` and update
   - `v :: [[e]]`
   - `concat :: [[e]] -> [e]`
4. apply `f = c` and update
   - `u :: [f]`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4

1. apply `[c] = a` and update
   - `u :: [c]`
2. apply `[[d]] = b` and update
   - `v :: [[d]]`
3. apply `[e] = [d]` to get `e = d` and update
   - `v :: [[e]]`
   - `concat :: [[e]] -> [e]`
4. apply `f = c` and update
   - `u :: [f]`
   - `head :: [f] -> f`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
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Given `f u v = min (head u) (last (concat v))`
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1. apply `f = e` and update
   - `v :: [[f]]`
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   - `last :: [[f]] -> [f]`

Type inference

Example (cont.)

Given `f u v = min (head u) (last (concat v))`
Goal `f :: Ord f => a -> b -> f`

Step 4 (cont.)

1. apply `f = e` and update
   - `v :: [[f]]`
   - `concat :: [[f]] -> [f]`
   - `last :: [[f]] -> [f]`
2. no further simplification possible,
   return `f :: Ord f => [f] -> [[f]] -> f`

Plan

Proofs

- Structural induction
- Case analysis
- Generalization
- Extensionality
- Computation induction

Structural induction

Induction on the structural definition of a datatype

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To prove property `P(x)` for all finite values `x` of type T,
prove `P(C)` for each constructor C of T.

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Induction on the structural definition of a datatype

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Induction on the structural definition of a datatype

To prove property `P(x)` for all finite values `x` of type T, prove `P(C)` for each constructor C of T.

- base cases are represented by proofs for non-recursive constructors
- inductive cases are represented by proofs for recursive constructors

Each recursive type parameter has a separate induction hypothesis. (Why?)

Structural induction on trees

Example

```
data Tree a = Leaf | Node (Tree a) a (Tree a)
```

Structural induction on trees

Example

```
data Tree a = Leaf | Node (Tree a) a (Tree a)

mirror Leaf = Leaf
mirror (Node l v r) = Node (mirror r) v (mirror l)

id x = x

(f . g) x = f (g x)
```

Structural induction on trees

Example

```
data Tree a = Leaf | Node (Tree a) a (Tree a)

mirror Leaf = Leaf
mirror (Node l v r) = Node (mirror r) v (mirror l)

id x = x

(f . g) x = f (g x)
```

Prove `(mirror . mirror) t .=. id t`.

Structural induction on trees

Example (cont.)

```
Lemma: (mirror . mirror) t .=. id t
```

Structural induction on trees

Example (cont.)

```
Lemma: (mirror . mirror) t .=. id t
Proof by induction on Tree t
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Proof by induction on Tree t
Case Leaf
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```
Lemma: (mirror . mirror) t .=. id t
Proof by induction on Tree t
Case Leaf
  To show: (mirror . mirror) Leaf .=. id Leaf
  Proof
                          (mirror . mirror) Leaf
```

Structural induction on trees

Example (cont.)

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Lemma: (mirror . mirror) t .=. id t
Proof by induction on Tree t
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  To show: (mirror . mirror) Leaf .=. id Leaf
  Proof
                                (mirror . mirror) Leaf
    (by def .)            .=. mirror (mirror Leaf)
```

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Example (cont.)

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Lemma: (mirror . mirror) t .=. id t
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  To show: (mirror . mirror) Leaf .=. id Leaf
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    (by def mirror)       .=. mirror Leaf
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    (by def mirror)       .=. Leaf
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Example (cont.)

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Lemma: (mirror . mirror) t .=. id t
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Case Leaf
  To show: (mirror . mirror) Leaf .=. id Leaf
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                                (mirror . mirror) Leaf
    (by def .)          .=. mirror (mirror Leaf)
    (by def mirror)     .=. mirror Leaf
    (by def mirror)     .=. Leaf
    (by def id)         .=. id Leaf
  QED
```

Structural induction on trees

Example (cont.)

```
Case Node l v r
```

Structural induction on trees

Example (cont.)

```
Case Node l v r
  To show: (mirror . mirror) (Node l v r)
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Example (cont.)

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  To show: (mirror . mirror) (Node l v r)
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  IH1:     (mirror . mirror) l .=. id l
  IH2:     (mirror . mirror) r .=. id r
  Proof
                          (mirror . mirror) (Node l v r)
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Example (cont.)

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  To show: (mirror . mirror) (Node l v r)
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    (by def mirror)
    .=. Node (mirror (mirror l)) v (mirror (mirror r))
    (by def .)
    .=. Node ((mirror . mirror) l) v (mirror (mirror r))
```

Structural induction on trees

Example (cont.)

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  To show: (mirror . mirror) (Node l v r)
           .=. id (Node l v r)
  IH1:     (mirror . mirror) l .=. id l
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    .=. Node ((mirror . mirror) l) v (mirror (mirror r))
    (by def .)
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Structural induction on trees

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  To show: (mirror . mirror) (Node l v r)
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    (by IH1)    .=. Node (id l) v ((mirror . mirror) r)
    (by IH2)    .=. Node (id l) v (id r)
    (by def id) .=. Node l v (id r)
    (by def id) .=. Node l v r
    (by def id) .=. id (Node l v r)
  QED
QED
```

Structural induction on lists

Definition of a list:

Structural induction on lists

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```
data [a] = [] | a : [a]
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Structural induction on lists

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To prove property `P(xs)` for all finite lists `xs`

- Base case: Prove `P([])`
- Inductive case: Prove `P(xs)` $\implies$ `P(x:xs)`

Structural induction on lists

Definition of a list:

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To prove property `P(xs)` for all finite lists `xs`

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Structural induction on lists
are inductions on the length of a list

Case analysis

For conditionals consider separate proofs for the cases `True` and `False`.

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`To show: if x < y then` A `else` B `.=. f x y`

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Example

```
To show: if x < y then A else B .=. f x y
Proof by case analysis on Bool x < y
Case True
  Assumption: x < y .=. True
  Proof
                          if x < y then A else B
```

Case analysis

For conditionals consider separate proofs for the cases `True` and `False`.

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To show: if x < y then A else B .=. f x y
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                          if x < y then A else B
    (by Assumption) .=. if True then A else B
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Proof by case analysis on Bool x < y
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  Assumption: x < y .=. True
  Proof
                                if x < y then A else B
    (by Assumption) .=. if True then A else B
    (by ifTrue)     .=. A
    ...
  QED
```

Case analysis

For conditionals consider separate proofs for the cases `True` and `False`.

Example

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Proof by case analysis on Bool x < y
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  Assumption: x < y .=. True
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                                if x < y then A else B
    (by Assumption) .=. if True then A else B
    (by ifTrue)     .=. A
    ...
  QED
Case False
  ...
QED
```

Generalization

When using the IH, variables may be replaced by arbitrary expressions, only the induction variable must stay fixed.

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Example

Consider a structural induction on `xs`
with the IH `f xs ys .=. g xs ys`.

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Example

Consider a structural induction on `xs`
with the IH `f xs ys .=. g xs ys`.
Then,

$$\texttt{f xs ys .=. g xs ys} \implies \texttt{f xs [] .=. g xs []}.$$

Generalization

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- a more generalized problem than the original problem; and
- that the specific instance of our problem follows from the generalized problem.

Extensionality

Two functions are equal
if for all arguments they yield the same result.

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if for all arguments they yield the same result.

Example

```
Lemma: f .=. g
Proof by extensionality with xs
  To show: f xs .=. g xs
  Proof by induction on List xs
    ...
  QED
QED
```

Computation induction

Induction on the length of a computation

Computation induction

Induction on the length of a computation

To prove property $\mathtt{P}(x_1, \ldots, x_k)$ for all $x_1, \ldots, x_k$,
for every defining equation

$$\mathtt{f}\ p_1, \ldots, p_k\ \mathtt{=}\ \ldots\ \mathtt{f}\ e_{11}, \ldots, e_{1k}\ \ldots\ \mathtt{f}\ e_{n1}, \ldots, e_{nk}\ \ldots$$

prove $\mathtt{P}(e_{11}, \ldots, e_{1k}), \ldots, \mathtt{P}(e_{n1}, \ldots, e_{nk}) \implies \mathtt{P}(p_1, \ldots, p_k)$.

Computation induction

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prove $\mathtt{P}(e_{11}, \ldots, e_{1k}), \ldots, \mathtt{P}(e_{n1}, \ldots, e_{nk}) \implies \mathtt{P}(p_1, \ldots, p_k)$.

Also referred to as an induction on the computation of a function `f` or `f`-induction.

Computation induction

Example

```
splice [] ys = ys
splice (x:xs) ys = x : splice ys xs
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Computation induction

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1. `P([], ys)`

Computation induction

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`splice`-induction: To prove `P(xs, ys)` for all `xs` and `ys`, prove

1. `P([], ys)`
2. `P(ys, xs)` $\Longrightarrow$ `P(x:xs, ys)`

Computation induction

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splice [] ys = ys
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`splice`-induction: To prove `P(xs, ys)` for all `xs` and `ys`, prove

1. `P([], ys)`
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Prove `length (splice xs ys) .=. length xs + length ys`.

Computation induction

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`splice`-induction: To prove `P(xs, ys)` for all `xs` and `ys`, prove

1. `P([], ys)`
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Prove `length (splice xs ys) .=. length xs + length ys`.

Structural induction does not work (why?)

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
Case 1
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
Case 1
  To show: length (splice [] ys) .=. length [] + length ys
  Proof
                          length (splice [] ys)
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
Case 1
  To show: length (splice [] ys) .=. length [] + length ys
  Proof
                          length (splice [] ys)
    (by def splice) .=. length ys

                          length [] + length ys
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
Case 1
  To show: length (splice [] ys) .=. length [] + length ys
  Proof
                                 length (splice [] ys)
    (by def splice) .=. length ys

                                 length [] + length ys
    (by def length) .=. 0 + length ys
```

Computation induction

Example (cont.)

```
Lemma: length (splice xs ys) .=. length xs + length ys
Proof by splice-induction on xs and ys
Case 1
  To show: length (splice [] ys) .=. length [] + length ys
  Proof
                                 length (splice [] ys)
    (by def splice) .=. length ys

                                 length [] + length ys
    (by def length) .=. 0 + length ys
    (by def 0)      .=. length ys
  QED
```

Computation induction

Example (cont.)

`Case 2`

Computation induction

Example (cont.)

```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
```

Computation induction

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```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
  IH:      length (splice ys xs)
           .=. length ys + length xs
  Proof
                          length (splice (x:xs) ys)
```

Computation induction

Example (cont.)

```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
  IH:      length (splice ys xs)
           .=. length ys + length xs
  Proof
                          length (splice (x:xs) ys)
    (by def splice) .=. length (x : splice ys xs)
```

Computation induction

Example (cont.)

```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
  IH:      length (splice ys xs)
           .=. length ys + length xs
  Proof
                          length (splice (x:xs) ys)
    (by def splice) .=. length (x : splice ys xs)
    (by def length) .=. 1 + length (splice ys xs)
```

Computation induction

Example (cont.)

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Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
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  Proof
                          length (splice (x:xs) ys)
    (by def splice) .=. length (x : splice ys xs)
    (by def length) .=. 1 + length (splice ys xs)
    (by IH)         .=. 1 + (length ys + length xs)
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Computation induction

Example (cont.)

```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
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                          length (splice (x:xs) ys)
    (by def splice) .=. length (x : splice ys xs)
    (by def length) .=. 1 + length (splice ys xs)
    (by IH)         .=. 1 + (length ys + length xs)
    (by comm_sum)   .=. 1 + (length xs + length ys)
```

Computation induction

Example (cont.)

Case 2

```
To show: length (splice (x:xs) ys)
         .=. length (x:xs) + length ys
IH:      length (splice ys xs)
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Proof
                          length (splice (x:xs) ys)
  (by def splice) .=. length (x : splice ys xs)
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  (by IH)         .=. 1 + (length ys + length xs)
  (by comm_sum)   .=. 1 + (length xs + length ys)
  (by assoc_sum)  .=. (1 + length xs) + length ys
```

Computation induction

Example (cont.)

```
Case 2
  To show: length (splice (x:xs) ys)
           .=. length (x:xs) + length ys
  IH:      length (splice ys xs)
           .=. length ys + length xs
  Proof
                                length (splice (x:xs) ys)
    (by def splice)  .=. length (x : splice ys xs)
    (by def length)  .=. 1 + length (splice ys xs)
    (by IH)          .=. 1 + (length ys + length xs)
    (by comm_sum)    .=. 1 + (length xs + length ys)
    (by assoc_sum)   .=. (1 + length xs) + length ys
    (by def length)  .=. length (x:xs) + length ys
  QED
QED
```

Structural vs computation induction

- structural induction
  inductive proof over the structural definition of a datatype.
- computation induction
  inductive proof over the structural definition of a function.

Plan

Correctness

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How can we prove that two modules implement the same structure?

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How can we prove that two modules implement the same structure?

$$\Longleftrightarrow$$

How can we prove that the implementation of one module simulates its counterpart?

Lists and sets

Each list $[x_1, \ldots, x_n]$ represents the set $\{x_1, \ldots, x_n\}$.

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Each list `[`x_1`, ...,` x_n`]` represents the set $\{x_1, \ldots, x_n\}$.
In mathematical terms:

α `::` `[a] -> {`a`}`
α `[`x_1`, ...,` x_n`] = {`x_1`, ...,` x_n`}`

α is an abstraction function.

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Lists simulate sets $\Longrightarrow$ α must be a homomorphism.

Lists and sets

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empty = []
insert x xs = if elem x xs then xs else x:xs
isin x xs = elem x xs
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```

Simulation requirements:

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`invar` must be preserved by every operation.

Correctness proof strategy

Let C and A be two modules that have the same interface: a type T and a set of functions F.

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  $\texttt{invar}\ x_1 \wedge \cdots \wedge \texttt{invar}\ x_n \implies \texttt{invar}\ (C.f\ x_1\ \ldots\ x_n)$
- $C.f$ simulates $A.f$
  $\texttt{invar}\ x_1 \wedge \cdots \wedge \texttt{invar}\ x_n \implies$
  $\alpha\ (C.f\ x_1\ \ldots\ x_n) = A.f\ (\alpha\ x_1)\ \ldots\ (\alpha\ x_n)$

Plan

I/O
- I/O in Haskell
- Sequencing
- Interlude: Monads

I/O

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To reason about programs like in mathematics, the programming language must have referential transparency. That is, any expression can be replaced by its value without changing the meaning of the program.

Programming languages that have referential transparency are called pure.

I/O in Haskell

Haskell distinguishes expressions without side effects (pure expressions) from expressions with side effects (actions) by their type:

```
IO a
```

is the type of (I/O) actions that return a value of type a.

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- `Char`: the type of pure expressions returning a `Char`

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Examples

- `Char`: the type of pure expressions returning a `Char`
- `IO Char`: the type of actions returning a `Char`
- `IO ()`: the type of actions returning nothing

`()` is the type of empty tuples with the only value `()`.

Basic actions

- `getChar :: IO Char`
  Reads a `Char` from standard input,
  echoes it to standard output,
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Basic actions

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  Reads a `Char` from standard input,
  echoes it to standard output,
  and returns it as the result
- `putChar :: Char -> IO ()`
  Writes a `Char` to standard output,
  and returns no result
- `return :: a -> IO a`
  Performs no action,
  just returns the given value as a result

Read/Show

- `Read`: parsing `String`

```
class Read a where
  read :: String -> a
```

Read/Show

- Read: parsing String

```
class Read a where
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- Show: converting to String

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class Show a where
  show :: a -> String
```

Important actions

- `putStr :: String -> IO ()`
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- `putStr :: String -> IO ()`
  Prints a string to standard output
- `putStrLn :: String -> IO ()`
  Prints a string followed by a newline to standard output
- `getLine :: IO String`
  Reads everything up until a newline from standard input

Sequencing

A sequence of actions can be combined into a single action with the keyword `do`.

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Example

```
get2 :: IO (Char,Char)
get2 = do x <- getChar    -- result is named x
          getChar         -- result is ignored
          y <- getChar
          return (x,y)
```

Sequencing

General format:

$$\begin{array}{ll} \texttt{do} & a_1 \\ & \vdots \\ & a_n \end{array}$$

Sequencing

General format:

```
do a1
   ⋮
   an
```

where each a_i can be one of

- an action
  Effect: execute action
- *x* `<-` *action*
  Effect: execute *action* `::` `IO` `a`, give result the name `x` `::` `a`
- `let` `x` `=` *expr*
  Effect: give *expr* the name `x`

Interlude: Monads

Monads are a general approach to computations that incur side effects.

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Interlude: Monads

Monads are a general approach to computations that incur side effects.

Idea: pipe data through the program implicitly.
In Haskell:

```
class Monad m where
  (>>=) :: m a -> (a -> m b) -> m b
  return :: a -> m a
```

```
do x <- act1
   act2
```

is syntactic sugar for

```
act1 >>= (\x -> act2)
```

Interlude: Monads

Example: `Maybe` as a monad

```
instance Monad Maybe where
  m >>= f = case m of
              Nothing -> Nothing
              Just x  -> f x
  return v = Just v
```

Interlude: Monads

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Using `do`, failure propagation and unwrapping of `Just` happens automatically.

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```
x :: Maybe Int
y :: Maybe Int
sum2 :: Maybe Int
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Interlude: Monads

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  return v = Just v
```

Using do, failure propagation and unwrapping of Just happens automatically.

```
x :: Maybe Int
y :: Maybe Int
sum2 :: Maybe Int
sum2 = do
  a <- someMaybeInt
  b <- anotherMaybeInt
  return (a + b)
```

Plan

Evaluation

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Expressions are evaluated (reduced) by successively applying definitions until no further reduction is possible.

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Expressions are evaluated (reduced) by successively applying definitions until no further reduction is possible.

An expression may have many reducible sub-expressions:

$$\underline{\texttt{sq}\ \underline{\texttt{(3+4)}}}$$

A reducible expression is also called redex.

Reduction strategies

- innermost reduces the innermost redex first

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  - corresponds to call by name

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- lazy combines an outermost reduction strategy with the sharing of expressions.

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  - corresponds to call by value
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  - unevaluated arguments are substituted into the function body
  - corresponds to call by name
- lazy combines an outermost reduction strategy with the sharing of expressions.
  - unevaluated arguments are substituted into the function body, but are only evaluated once for all copies of the same expression

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- outermost reduces the outermost redex first
  - unevaluated arguments are substituted into the function body
  - corresponds to call by name
- lazy combines an outermost reduction strategy with the sharing of expressions.
  - unevaluated arguments are substituted into the function body, but are only evaluated once for all copies of the same expression
  - *call by need*

Theorems

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  $\Longrightarrow$ outermost reduction terminates as often as possible
- Lazy evaluation never needs more steps than innermost reduction.

Principles of lazy evaluation

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Why?

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- Arguments are not necessarily evaluated fully, but only far enough to evaluate the function.
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Haskell never reduces inside a lambda

Why?

- lazy evaluation uses as few steps as possible
- functions can only be applied

Infinite lists

Example: `head ones`

```
ones :: [Int]
ones = 1 : ones
```

`ones` defines an infinite list of 1s. `ones` is called a producer.

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Innermost reduction:

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= ...
```

Infinite lists

Haskell lists are never actually infinite
but only potentially infinite

Lazy evaluation computes as much of the infinite list as needed

Plan

Time complexity analysis

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Calculating $T_f(n)$:

1. from the equations for f derive equations for T_f
2. if the equations for T_f are recursive, solve them

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(x:xs) ++ ys = x : (xs ++ ys)
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$T_{++}(0, n) = O(1)$
$T_{++}(m + 1, n) = T_{++}(m, n) + O(1)$
$\implies T_{++}(m, n) = O(m)$